

2-Mark Questions & Answers — Data Structures Module 2: Stacks & Queues

Q1. Define Stack ADT and list its primary operations.

A **Stack** is a linear data structure following the **Last-In, First-Out (LIFO)** principle.

- **Operations:** `push(x)` (insert at top), `pop()` (delete from top), `peek()` / `top()` (inspect top element), `isEmpty()`, `isFull()`.

Q2. What is Stack Overflow and Stack Underflow?

- **Stack Overflow:** Attempting to `push` an element onto a completely filled stack (`top == MAX - 1`).
- **Stack Underflow:** Attempting to `pop` or `peek` from an empty stack (`top == -1`).

Q3. Define Queue ADT and state FIFO principle.

A **Queue** is a linear data structure following the **First-In, First-Out (FIFO)** principle, where elements are inserted at the **Rear** end and deleted from the **Front** end.

Q4. What is the advantage of a Circular Queue over a Linear Queue?

In a Linear Queue, space freed by `dequeue` cannot be reused once `rear == MAX - 1` (false overflow). A **Circular Queue** wraps pointers around using modulo arithmetic (`(rear + 1) % MAX`), eliminating memory wastage.

Q5. What is a Double-Ended Queue (Deque)?

A **Deque** is a generalized queue where insertion and deletion operations can be performed efficiently from **both ends** (Front and Rear).